Prospects for constraining angular momentum transport in solar-like MS stars with PLATO

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In prep.

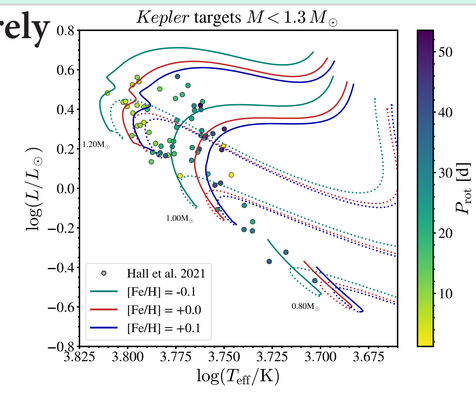
We assess the prospects for constraining different models of AM transport in solar-like MS stars with the upcoming ESA PLATO mission. We test different methods that combine asteroseismic measurements of the average internal rotation with surface rotation measurements from spot modulation of the stellar lightcurve.

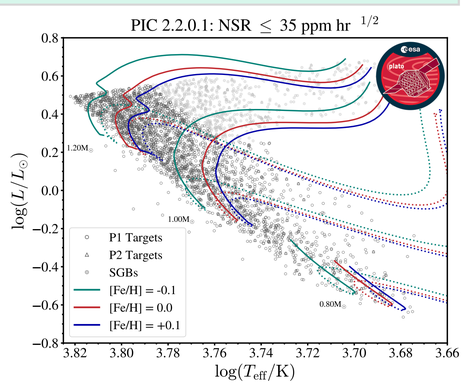
MOTIVATION

AM transport remains **poorly constrained in solar-like MS stars:** current observations **rule out purely hydrodynamic transport** for the Sun and some cool stars, but provide **limited constraints on more efficient mechanisms.**

Rotational splittings has been measured for ≲100 *Kepler* [1] solar-like MS stars. **PLATO [2]** is expected to **increase this sample by ~2 orders of magnitude, with comparable or better S/N and ≥2-yr light curves [3].**

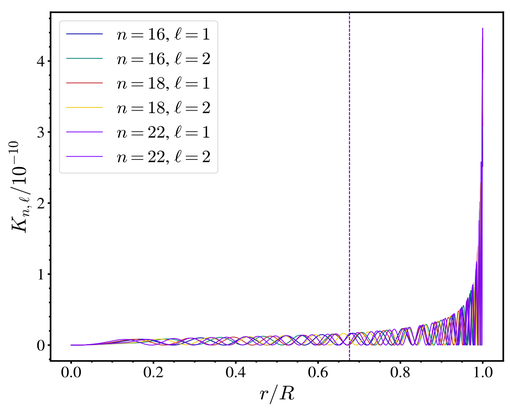
PLATO is expected to **better preserve long-timescale variability [4], improving surface-rotation measurements at** P≳30 d, where *Kepler* is increasingly affected by systematics.





CHALLENGES

p-modes rotational splittings gives only a **weighted average** of the rotation profile **dominated by the surface layers**

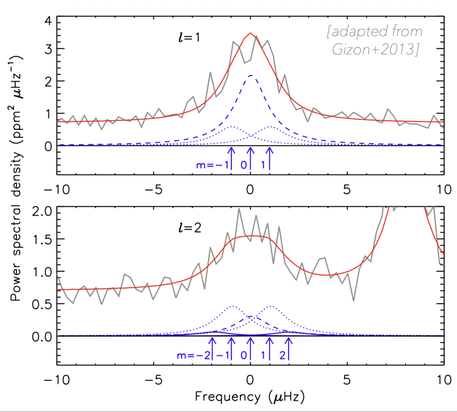


Rotational splitting

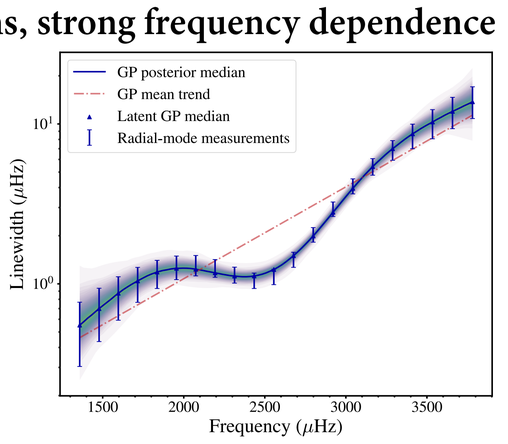
$$\delta\omega_{nlm} = m\beta_{nl}\int_0^R K_{nl}(r)\Omega(r)\mathrm{d}r$$

Different modes have similar K_{nl}
↓
Similar $\delta\omega_{nlm}$
↓
Rotation profile cannot be resolved

- Usually **slow rotators** → rotational **splittings are small**
- Modes **lifetimes are short** → **broad linewidths, strong frequency dependence**



We assume:
- $\Gamma_{nlm} = \Gamma_{n0}$
- Γ_{n0} correlated in frequency through a latent GP with parameters sampled in the MCMC



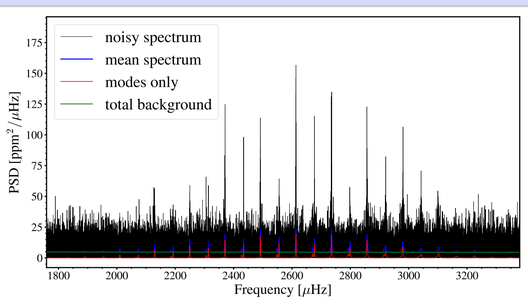
STELLAR MODELS, ROTATIONAL SPLITTINGS AND SYNTHETIC POWER SPECTRA

Rotating stellar models: MESA [5]
- different AM transport processes,
- surface magnetic braking (& WMB),
- different initial periods.

Computed oscillation modes:
- frequencies,
- eigenfunctions,
- rotational splittings.

GYRE [6]

Computed synthetic PSDs:
- modes properties calibrated on Kepler observations.



AVERAGE ASTEROSEISMIC ROTATION VS SURFACE ROTATION

I extracted the **average rotational splitting** across al visible modes to increase precision (splitting expected to be very similar across modes). The average asteroseismic rotation is just $\langle\delta\omega_{nlm}/m\beta_{nl}\rangle$:

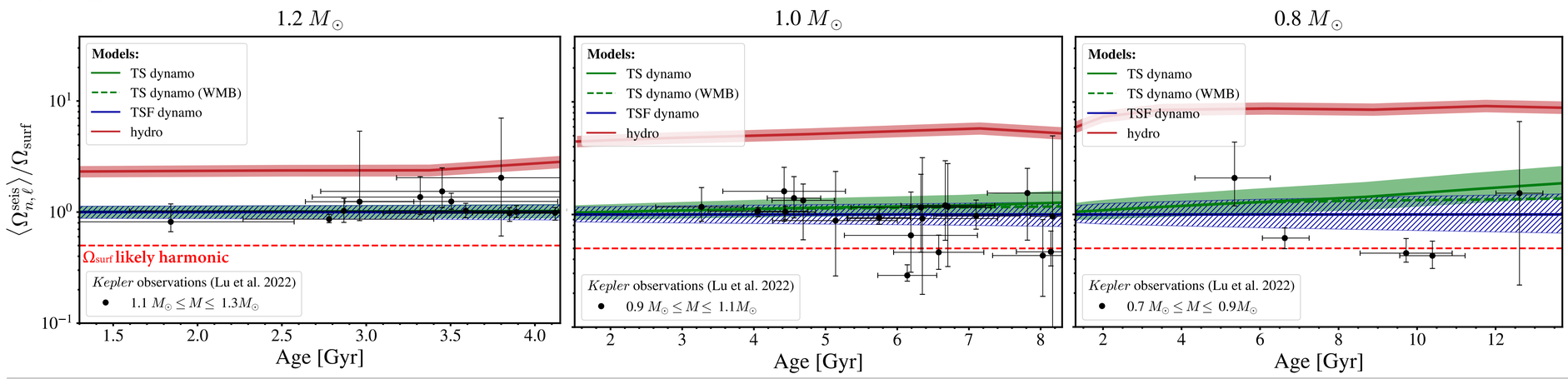
$$\Omega^{\rm seis}_{n,\ell} = \int_0^R K_{nl}(r)\Omega(r)\mathrm{d}r \;\rightarrow\; \langle\Omega^{\rm seis}_{n,\ell}\rangle/\Omega^{\rm surf} > 1$$

Interior rotates faster than $\Omega_{\rm surf}$

Uncertainties on $\delta\omega_{nlm}$ comes from peakbagging the synthetic PSD

Uncertanties on the surface rotation are taken from the simulations of [4].

NSR = 20 ppm hr $^{1/2}$
$T_{\rm obs}$ = 2yr



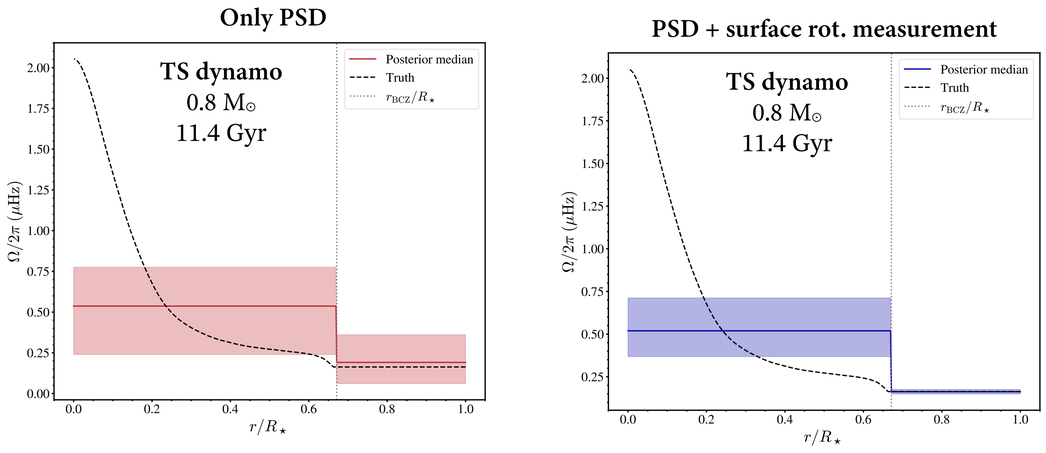
Pure hydro models clearly distinguishable from solid body rotation (already ruled out by Kepler observations).

Tayler-Spruit (TS) dynamo [7] models (magnetic instability) with M≲0.9M⊙ distinguishable from solid body rotation.

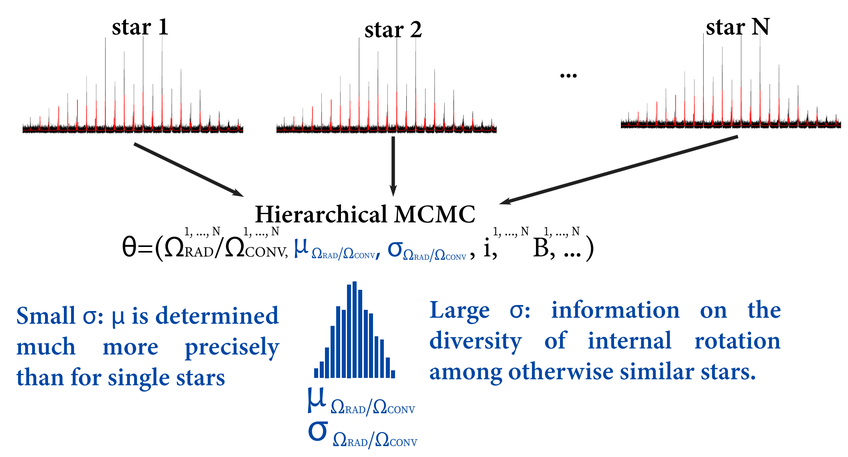
Tayler-Spruit-Fuller (TSF) dynamo [8] models (revised TS prescription) remains always consistent with solid body rotation.

MCMC FITTING OF THE FULL PSD WITH A PARAMETRIC ROTATION PROFILE AND SURFACE CONSTRAINT

MCMC fitting of the full PSD, using a **parametric rotation profile** provides **improved constraints** when combined with a **surface rotation measurement.**



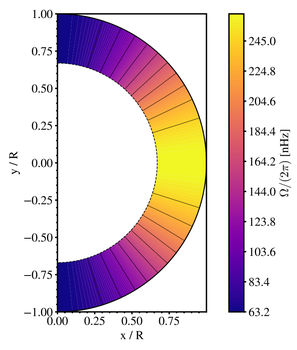
We are extending this framework to **hierarchical MCMC**, leveraging PLATO's large sample to **strengthen constraints and enable population-level studies:**



Small σ: μ is determined much more precisely than for single stars

Large σ: information on the diversity of internal rotation among otherwise similar stars.

Similar approach for $\Delta\Omega_{\rm LAT}$: $\Delta\Omega_{\rm LAT}=\Omega_{\rm POLE}-\Omega_{\rm EQ}$



CONCLUSIONS

PLATO's large, high-quality sample of solar-like MS stars will provide stronger constraints on AM transport. In particular, M≲0.9M⊙ targets are expected to constrain TS dynamo models. Joint MCMC fitting of the full PSD and surface rotation can further constrain parametric internal-rotation profiles. Extending this framework to hierarchical inference can constrain the population distribution of radial and latitudinal differential rotation, particularly for populations with small intrinsic scatter.

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